

Occupational Health Risk Assessment - Viaduct Works

Document no: 1MC05-ALJ-HS-RIA-.....(to be completed by ALIGN)

Work Package Ref:

One tab per sub-contractor

Revision	Author	Reviewed by	Approved by	Date	Revision Details
C01	Park Health	H&S Manager	Work package manager	11/05/2020	First issue with initial assessment and sub-contractor input.

STAKEHOLDER REVIEW REQUIRED (SRR)

Purpose of SRR

☐ COUNTY/DISTRICT/LONDON BOROUGH COUNCIL

☐ LOV

☐ LUL

☐ NRL

☐ TFL

☐ UTILITIES COMPANY

☐ OTHER

☐ ACCEPTANCE

☐ APPROVAL

☐ NO OBJECTION

☐ CONSENT

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SCOPE OF WORKS AND CONSIDERATIONS: The works will include but not limited to the following detailed design, supply and installation. Design of works is being completed by Align and Align-D, civil package of works is being completed by Roadbridge.																																			
Details of activities related to the Location / Scope of Works: -Site Establishments and Compounds -Utilities Works -Highway Works -Jetty Construction -Cofferdam Construction -General Working over water																																			
This HRA will be used in accordance with the Align Health Management plan - 1MC05-ALJ-HS-PLN-C001-000003.																																			
Contractors must review and complete the HRA, confirming or justifying risk calculations / control measures and adding any health risks they believe are not captured in this initial assessment.																																			
HOLD POINT: Minimum controls have been stipulated but the HRA will be updated once the method statements are received and shall be updated as often as necessary.																																			
Ref	Agent of concern		Health Hazard	Work location	Degree of Risk			Control Measures	Potential Exposure monitoring needs			Potential Health Surveillance needs						Potential training needs	Confirmation of risk. Does it exist Yes/NO if No give justification	Actual Control Measures to be used	Degree of residual risk			Actual Exposure monitoring needs		Actual Health Surveillance needs						Actual training needs			
					L	S	R		Noise	HAVS	Air	Audio	Baseline			Routine					L	S	R	Noise	HAVS	Air	Audio	Baseline			Routine				
													HAVS	Skin	Lungs	HAVS	Skin											Lungs	HAVS	Skin	Lungs		HAVS	Skin	Lungs
Site Establishment and Compound																																			
	Ground contamination from soils		Potential exposure to lead and other heavy metals: headaches, stomach pains, anemia, kidney damage, nerve and brain damage and infertility Hydrocarbon exposure may cause dermatitis, vapours may drowsiness and dizziness	Compound Setup - Breaking ground	1	2	2	Refer to specific COSHH risk assessment for the materials used. Wear rubber/plastic gloves during use. Clean hands after any exposure. Stockpile management - Cover piles and damp down																											
	Asbestos contamination within soils		Asbestosis, mesothelioma, lung cancer	Compound Setup - Breaking ground	1	4	4	Asbestos in soils procedure. Workforce to undertake watching brief and stop works on the discovery of suspected asbestos containing materials										Asbestos Awareness Training																	
	Dust	Wood	Lung disorders as a result of breathing in wood dust. Cancer, asthma, irritation	Compound Setup - Shuttering and other construction activities	2	3	6	On tool bags used in conjunction with P3 RPE Face fit testing for all RPE, or on tool extraction used.																											
	Gases/Fumes	Diesel Engine Exhaust Emissions (DEEE)	Class 1 human carcinogen. Acute irritation. Drowsiness and dizziness	Compound Setup - use of plant	1	4	4	Diesel plant to be fitted with filtration/catalytic convertor systems (DPFs and AdBlue). Keep diesel-powered equipment exhausts away from potential confined spaces. Personnel – vehicle segregation during HGV operations																											
	Noise	Hand Tools	Noise induced hearing loss, Tinnitus, Threshold Shift	Compound Setup - Shuttering and other construction activities	3	3	9	Low noise emitting tools to be used. Noise screens to be used around operation. Wear hearing protection when noise levels exceed 80dB(A). Wear hearing protection when using impact wrenches	✓			✓			✓			Noise Exposure TBT																	
	Noise	Plant Operations	Noise induced hearing loss, Tinnitus, Threshold Shift	Compound Setup - Normal construction activities	1	3	3	Wear hearing protection when noise levels exceeds 80dB(A) Segregation of plant from operatives. Plant doors and windows to be closed. Hearing protection zones.																											
	Vibration (HAV)	Hand Tools.	Hand Arm Vibration Syndrome, vibration white finger, nerve damage	Compound Setup - Shuttering and other construction activities	3	2	6	Selection of low vibration equipment Well-maintained equipment. HAVS awareness training. Use gloves to keep hands warm and dry		✓		✓			✓			HAVS Awareness Training																	
	Vibration (WBV)	From Plant	Lower back pain, nausea	Normal construction activities	1	2	2	Ensure seats are fully padded and adjustable for the users weight.																											
	Wet concrete		Dermatitis/ concrete burns/ sensitisation	Compound Setup - Slab pouring	3	3	9	Ensure waterproof and high pH resistant gloves. Concrete overalls are worn to keep all skin covered. Wellington boots are worn. Follow requirements stated on the COSHH assessment. Wash PPE and skin after work.					✓			✓		Dermatitis TBT, if likelihood increases																	
	Ergonomics - Manual handling		Musculoskeletal disorders	Compound Setup - Shuttering and other construction activities	3	3	9	Use of lifting aids where possible. Team Lifting Ensure clear site and good ground conditions to promote good handling practices. Ensure operatives are not reaching too far as well as not repetitive twisting.										Manual handling training																	
Utilities Works																																			
	Ground contamination from soils		Potential exposure to lead and other heavy metals: headaches, stomach pains, anemia, kidney damage, nerve and brain damage and infertility Hydrocarbon exposure may cause dermatitis, vapours may drowsiness and dizziness	Excavation of unknown ground	2	3	6	Refer to specific COSHH risk assessment for the materials used. Wear rubber/plastic gloves during use. Clean hands after any exposure. Stock pile management - Cover piles and damp down																											
	Noise	Plant operations	Noise induced hearing loss, Tinnitus, Threshold Shift	Exposure during excavation and leveling	3	3	9	Wear hearing protection when noise levels exceeds 80dB(A). Segregation of plant from operatives. Plant doors and windows to be closed. Hearing protection zones.	✓			✓			✓			Noise Exposure TBT																	
	Ergonomics - Manual handling		Musculoskeletal disorders	Manipulation, placing and removal of utilities assets	3	3	9	Use of lifting aids and levers where possible. Manipulation in teams. Mininimse manipulation from MEWPS Ensure clear site and good ground conditions to promote good handling practices. Ensure good working positions to allow to operatives to not reach too far as well as not repetitive twisting.										Manual handling training																	

Ref	Agent of concern		Health Hazard	Work location	Degree of Risk			Control Measures	Potential Exposure monitoring needs			Potential Health Surveillance needs				Potential training needs	Confirmation of risk. Does it exist Yes/NO if No give justification	Actual Control Measures to be used	Degree of residual risk			Actual Exposure monitoring needs			Actual Health Surveillance needs				Actual training needs					
					L	S	R		Noise	HAVS	Air	Baseline			Routine				L	S	R	Noise	HAVS	Air	Baseline			Routine						
												Audio	HAVS	Skin	Lungs										Audio	HAVS	Skin	Lungs		Audio	HAVS	Skin	Lungs	
	Wet concrete		Dermatitis/ concrete burns/ Sensitisation	Construction of new utilities assets	2	3	6	Ensure waterproof and high pH resistant gloves. Concrete overalls are worn to keep all skin covered. Wellington boots are worn. Follow requirements stated on the COSHH assessment. Wash PPE and skin after work.						✓			✓	Dermatitis TBT, if likelihood increases																
	Noise	Hand Tools	Noise induced hearing loss, Tinnitus, Threshold Shift	Breaking ground near utilities	2	3	6	Low noise emitting tools to be used. Wear hearing protection when noise levels exceed 80dB(A). Wear hearing protection when using impact wrenches	✓			✓			✓			Noise Exposure TBT																
	Vibration (HAV)	Hand Tools.	Hand Arm Vibration Syndrome, vibration white finger, nerve damage	Breaking ground near utilities	4	3	12	Selection of low vibration equipment Well-maintained equipment. HAVS awareness training. Use gloves to keep hands warm and dry	✓			✓			✓			HAVS Awareness Training																
	Dust	Cement/ Concrete	Lung disorders as a result of breathing in cement / grout dust. Lung cancer and silicosis	Drilling / manipulating and altering PC panels	2	3	6	On tool extraction or damping down used. if high exposures used with conjunction with P3 RPE. Face fit testing for all RPE.			✓			✓			✓	Lung health TBT																
Highway Works																																		
	Noise	Plant operations	Noise induced hearing loss, Tinnitus, Threshold Shift	Exposure during excavation and leveling	3	3	9	Wear hearing protection when noise levels exceeds 80dB(A). Segregation of plant from operatives. Plant doors and windows to be closed. Hearing protection zones.	✓			✓			✓			Noise Exposure TBT																
	Dust	Cement/ Concrete	Lung disorders as a result of breathing in cement / grout dust. Lung cancer and silicosis	Base aggregate delivery and spreading	2	3	6	Specify wet/damp delivery. Dampen aggregate where required ahead of manipulation. For high exposures use in conjunction with P3 RPE. Face fit testing for all RPE.			✓			✓			✓	Lung health TBT																
	Vibration (WBV)	Ride on rollers	Lower back pain, nausea	Exposure during roller operations	3	2	6	Ensure seats are fully padded and adjustable for the users weight																										
	Noise	Plant operations	Noise induced hearing loss, Tinnitus, Threshold Shift	Exposure during excavation and leveling	3	3	9	Wear hearing protection when noise levels exceeds 80dB(A). Segregation of plant from operatives. Plant doors and windows to be closed. Hearing protection zones.	✓			✓			✓			Noise Exposure TBT																
	Gases/Fumes	Diesel Engine Exhaust Emissions (DEEE)	Class 1 human carcinogen. Acute irritation. Drowsiness and dizziness	Exposure during plant operations	1	4	4	Diesel plant to be fitted with filtration/catalytic convertor systems (DPFs and AdBlue). Keep diesel-powered equipment exhausts away from potential confined spaces. Ensure good supply of air into the shaft during sinking Personnel – vehicle segregation during HGV operations																										
	Wet concrete		Dermatitis/ concrete burns/ sensitisation	Kerbstone and road furniture laying	2	3	6	Ensure waterproof and high pH resistant gloves. Concrete overalls are worn to keep all skin covered. Wellington boots are worn. Follow requirements stated on the COSHH assessment. Wash PPE and skin after work.						✓			✓	Dermatitis TBT, if likelihood increases																
	Dust	Cement/ Concrete	Lung disorders as a result of breathing in cement / grout dust. Lung cancer and silicosis	Cutting kerbstones	2	3	6	On tool extraction or damping down used. if high exposures used with conjunction with P3 RPE. Face fit testing for all RPE.			✓			✓			✓	Lung health TBT																
	Ergonomics - Manual handling		Musculoskeletal disorders	Lifting and placing kerbstones and road furniture	3	3	9	Use of lifting aids and levers where possible. Manipulation in teams. Minimise manipulation from MEWPS. Ensure clear site and good ground conditions to promote good handling practices. Ensure good working positions to allow to operatives to not reach too far as well as not repetitive twisting.										Manual handling training																
	COSHH - Direct contact	Bitumen	Dermatitis, allergic reactions, skin irritation. Some evidence of carcinogenic effect	Black topping	3	3	9	Refer to specific COSHH risk assessment for the materials used. Wear rubber/plastic gloves during use. Clean hands after any exposure.					✓				✓	Skin self check training																
Jetty Construction																																		
	Noise	Piling operations	Noise induced hearing loss, Tinnitus, Threshold Shift	Exposure during tubular piling operations	2	3	6	Wear hearing protection when noise levels exceeds 80dB(A). Segregation of plant from operatives. Plant doors and windows to be closed. Hearing protection zones.	✓			✓			✓			Noise Exposure TBT																
	COSHH - Direct contact	Greases and lubricants	Dermatitis, allergic reactions, skin irritation	Exposure during tubular piling operations	2	2	4	Refer to specific COSHH risk assessment for the materials used. Wear rubber/plastic gloves during use. Clean hands after any exposure.						✓			✓																	
	Vibration (WBV)	From Plant	Lower back pain, nausea	Exposure during CFA piling operations	1	2	2	Ensure seats are fully padded and adjustable for the users weight																										
	Ergonomics - Manual handling		Musculoskeletal disorders	Manipulating tubular piles into position	3	3	9	Use of lifting aids and levers where possible. Manipulation in teams. Ensure clear site and good ground conditions to promote good handling practices. Ensure good working positions to allow to operatives to not reach too far as well as not repetitive twisting.										Manual handling training																
	Noise	Hand Tools	Noise induced hearing loss, Tinnitus, Threshold Shift	Fixing steel works	3	3	9	Low noise emitting tools to be used. Noise screens to be used around operation. Wear hearing protection when noise levels exceed 80dB(A). Wear hearing protection when using impact wrenches	✓			✓			✓			Noise Exposure TBT																
	Vibration (HAV)	Hand Tools.	Hand Arm Vibration Syndrome, vibration white finger, nerve damage	Fixing steel works	3	2	6	Selection of low vibration equipment Well-maintained equipment. HAVS awareness training. Use gloves to keep hands warm and dry	✓			✓			✓			HAVS Awareness Training																
	Ergonomics - Manual handling		Musculoskeletal disorders	Manipulating steels and PR elements into position	3	3	9	Use of lifting aids and levers where possible. Manipulation in teams. Minimise manipulation from MEWPS. Ensure clear site and good ground conditions to promote good handling practices. Ensure good working positions to allow to operatives to not reach too far as well as not repetitive twisting.										Manual handling training																

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					L	S	R		Noise	HAVS	Air	Audio	HAVS	Skin	Lungs	Audio	HAVS				Skin	Lungs														
	Dust	Cement/ Concrete	Lung disorders as a result of breathing in cement / grout dust. Lung cancer and silicosis	Drilling / manipulating and altering PC panels	2	3	6	On tool extraction or damping down used, if high exposures used with conjunction with P3 RPE. Face fit testing for all RPE.			✓				✓		✓	Lung health TBT																		
	Noise	Hand Tools	Noise induced hearing loss, Tinnitus, Threshold Shift	Drilling / manipulating and altering PC panels	2	3	9	Low noise emitting tools to be used. Wear hearing protection when noise levels exceed 80dB(A). Wear hearing protection when using impact wrenches	✓			✓			✓			Noise Exposure TBT																		
	Vibration (HAV)	Hand Tools.	Hand Arm Vibration Syndrome, vibration white finger, nerve damage	Drilling / manipulating and altering PC panels	2	2	6	Selection of low vibration equipment Well-maintained equipment. HAVS awareness training. Use gloves to keep hands warm and dry	✓			✓			✓			HAVS Awareness Training																		
Cofferdam Construction																																				
	Noise	Piling operations	Noise induced hearing loss, Tinnitus, Threshold Shift	Exposure during sheet piling operations	3	3	6	Wear hearing protection when noise levels . Segregation of plant from operatives. Plant doors and windows to be closed. Hearing protection zones.	✓			✓			✓			Noise Exposure TBT																		
	Vibration (WBV)	From Plant	Lower back pain, nausea	Exposure during sheet piling operations	3	2	6	Ensure seats are fully padded and adjustable for the users weight.																												
	Vibration (HAV)	From Plant	Hand Arm Vibration Syndrome, vibration white finger, nerve damage	Exposure to rig operator through joysticks	2	2	4	Selection of low vibration equipment Well-maintained equipment. HAVS awareness training. Use gloves to keep hands warm and dry																												
	Ergonomics - Manual handling		Musculoskeletal disorders	Manipulating sheet piles into position	3	3	9	Use of lifting aids and levers where possible. Manipulation in teams. Minimise manipulation from MEWPS. Ensure clear site and good ground conditions to promote good handling practices. Ensure good working positions to allow to operatives to not reach too far as well as not repetitive twisting.										Manual handling training																		
	Noise	Plant operations	Noise induced hearing loss, Tinnitus, Threshold Shift	Exposure during cofferdam excavation	3	3	9	Wear hearing protection when noise levels exceeds 80dB(A). Segregation of plant from operatives. Plant doors and windows to be closed. Hearing protection zones.	✓			✓			✓			Noise Exposure TBT																		
	Noise	Hand Tools	Noise induced hearing loss, Tinnitus, Threshold Shift	Fixing temporary proping	3	3	9	Low noise emitting tools to be used. Noise screens to be used around operation. Wear hearing protection when noise levels exceed 80dB(A). Wear hearing protection when using impact wrenches	✓			✓			✓			Noise Exposure TBT																		
	Vibration (HAV)	Hand Tools.	Hand Arm Vibration Syndrome, vibration white finger, nerve damage	Fixing temporary proping	3	2	6	Selection of low vibration equipment Well-maintained equipment. HAVS awareness training. Use gloves to keep hands warm and dry	✓			✓			✓			HAVS Awareness Training																		
	Gases/Fumes	Welding fumes	Occupational asthma. Lung cancer. Irritation of lungs	Fixing temporary proping. Sealing of the cofferdam	3	3	9	Local exhaust ventilation in conjunction with RPE Face fit testing for all close-fitting RPE						✓			✓																			
	COSHH - Direct contact and vapour exposure	Sealing Cofferdam piles	Dermatitis, allergic reactions, skin irritation, lung irritation, asthma	Waterproofing shet piles	3	3	9	Refer to specific COSHH risk assessment for the materials used. Wear rubber/plastic gloves during use. Clean hands after any exposure.					✓				✓																			
Working over and adjacent to water																																				
	Vibration (HAV)	Outboard motor	Hand Arm Vibration Syndrome, vibration white finger, nerve damage	Whilst manning the support boat	3	2	6	HAVS awareness training. Use gloves to keep hands warm and dry		✓			✓			✓		HAVS Awareness Training																		
	Noise	Outboard motor	Noise induced hearing loss, Tinnitus, Threshold Shift	Whilst manning the support boat	3	3	9	Wear hearing protection when noise levels exceed 80dB(A). Wear hearing protection when using impact wrenches	✓			✓			✓			Noise Exposure TBT																		
	COSHH Biological exposures		Enterococci, E.coli and coliform bacteria: diarrhoea, gastroenteritis Leptospirosis leading to Wells disease	Contact with water courses during construction	2	3	6	Protect open wounds with waterproof bandages. Waterproof gloves and clothing protection. Personal hygiene – wash stations available, washing hands before all meals and rest breaks and after all sampling activities										Waterbourne illness/Wells disease awareness training																		
	COSHH Biological exposures		Enterococci, E.coli and coliform bacteria: diarrhoea, gastroenteritis Leptospirosis leading to Wells disease	Contact with water courses during manning support boat	3	3	9	Protect open wounds with waterproof bandages. Waterproof gloves and clothing protection. Personal hygiene – wash stations available, washing hands before all meals and rest breaks and after all sampling activities										Waterbourne illness/Wells disease awareness training																		

		Exposure rating (Likelihood)				
		1	2	3	4	5
Consequence of Health Hazard Exposure (Severity)		Never heard of in the setting	Related incidents / Exposures in similar setting	Incident / Exposure has occurred on site or within similar setting	Incident / exposure occurs several times per year in organisation	Incident / exposure occurs several times per year in location
1	Slight injury or temporary health effect	1	2	3	4	5
2	Minor injury or health effect	2	4	6	8	10
3	Major injury or major health effect	3	6	9	12	15
4	Disablement or Debilitating Health Effect	4	8	12	16	20
5	Fatality (including Multiple) from exposure or occupational illness	5	10	15	20	25

LOW Health Hazard Risk Rating in setting
MEDIUM Health Hazard Risk Rating in setting
HIGH Health Hazard Risk Rating in setting